

2. Explain the difference between method declaration and method body.

The method declaration is the first line of the method. It includes access level, return type, name and parameters. The method body contains the statements that implement the method.

3. What type of keyword is used to change the access level of a method?

Access modifiers: public, private, protected.

4. What is another word used for describing the access level of a method?

Visibility

5. Explain the scope of each of the variables in the code below:

```
public class ScopeExample {  
    public static void main(String[] args) {  
        int var1;  
        for (int var2 = 0; var2 < 5; var2++) {  
            method1();  
        }  
    }  
}
```

```
public static void method1() {  
    int var3;  
    for (int var4 = 0; var4 < 2; var4++) {  
        var3 += 1;  
    }  
}
```

var1 is declared and limited inside of the main() method.

var2 is declared and limited inside of the for loop in the main method.

var3 is declared in the method1 and limited inside of the method1().

var4 is declared and limited inside of the for loop in the method1().

6. Write a method declaration for each of the following descriptions:

a) A class method named getVowels that can be called by any other method, requires a String parameter, and returns an integer value.

```
Public static int getVowels(String str){  
}
```

b) A class method named extractDigit that can be called by any other method, requires an integer parameter, and returns an integer value.

```
Public static int extractDigit(int num){  
}
```

c) A class method named insertString that can be called by any other method, requires a String parameter and an integer parameter, and returns a String parameter.

```
Public static String insertString(String str, int num){  
}
```

7. a) How does the compiler distinguish one method from another?

They can be differentiated as long as they have different parameters lists. They have different parameter types, order and number.

b) Can two methods in the same class have the same name? Explain.

Yes, it's going to be method overloading, they can be distinguished by the method signature. (Cannot differentiate them by looking at the return type).

8. a) What is the return statement used for?

To give a value back to the user by using an exit method.

b) How many values can a return statement send back to the calling statement?

One.

c) How is the declaration of a method returning a value different from the declaration of a method that does not return a value?

Methods that return a value must declare a specific type and must return a value of that type, while methods that do not return a value of that type, declared with void and do not return any value.

9. The errors in this code

11. a) T

b) F A method call does not need to be in an assignment statement, it can be alone.

c) F The void method cannot return a value.

d) F The return type comes after it.

e) T

f) F Method parameters are enclosed by ()

g) F it only can be used when they are inside of the same class.

h) T

i) F Method overloading means that within one same class there are different methods with the same name.

j) T

k) F A precondition states what must be true before a method runs, not the data types of parameters.

l) T